

IoT Platforms

Frank Walsh

What are IoT platforms

- IoT applications combine sensors, devices, data, analytics and integrations in a seamless and unified way
 - e.g. your project!
- IoT Platforms provide software tools and components to:
 - connect sensors, devices, and data networks
 - Analyse and store data
 - Integrate with other apps
- So what? We know the tech for that now (I2C, SPI, BLE, MQTT, Python...)
- Main selling point of an IoT platform is software that it
 - accelerates the IoT development process
 - Focuses on IoT: brings in best of breed features
 - Provides initial scaffolding for IoT projects

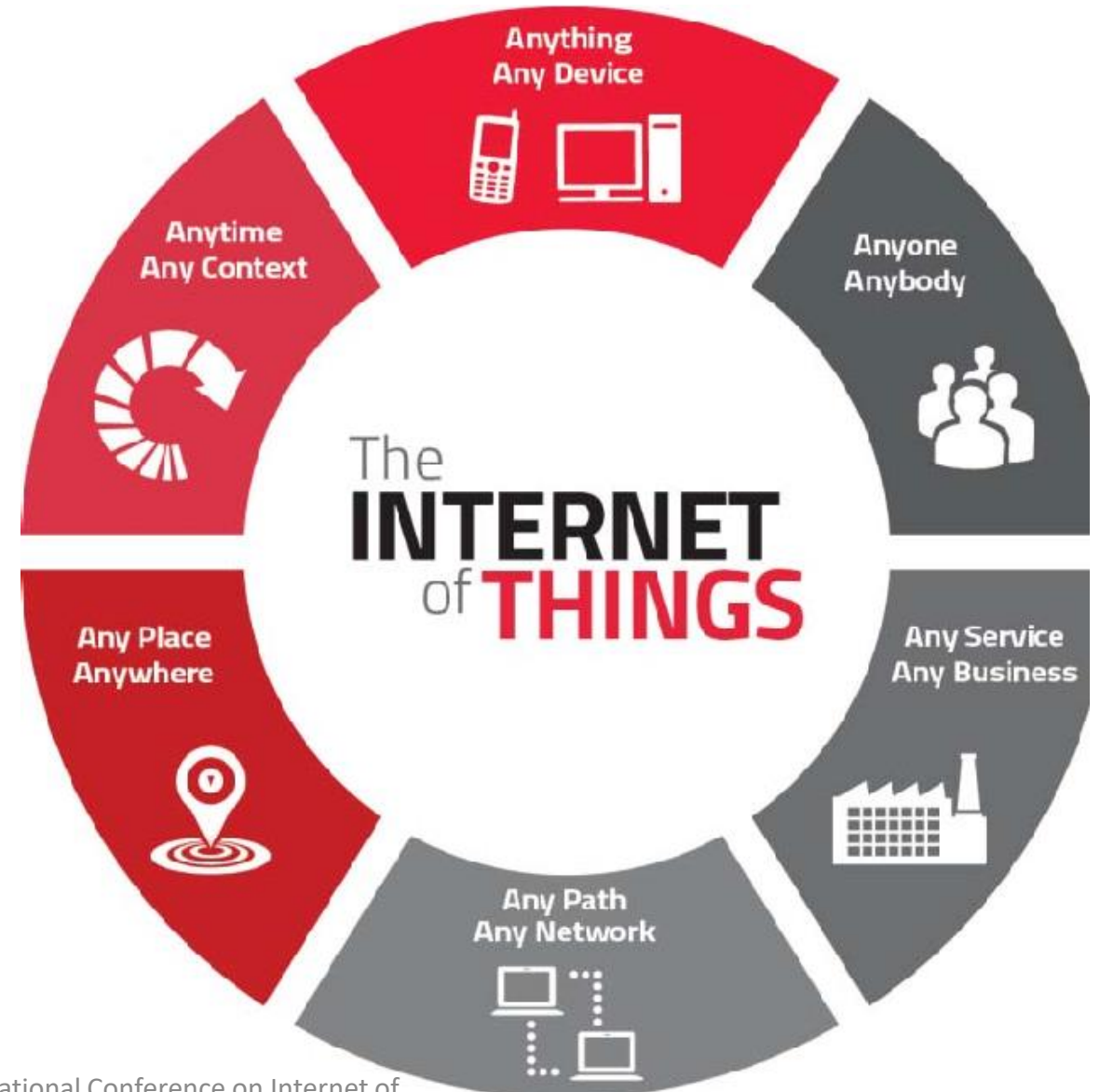
What are IoT Platform



- Many(not all) are cloud-based platforms that require subscription
- Provide device/language agnostic set of Software Development kits
 - Arduino/RPi/beagleboard
- IoT development is generally iterative:
 - Starts with initial simple use case
 - Once operational, data/insights result in new usecases
- IoT platforms should promote scalable, iterative development
 - Allow for quick app development
 - Ability to adapt/optimize apps quickly

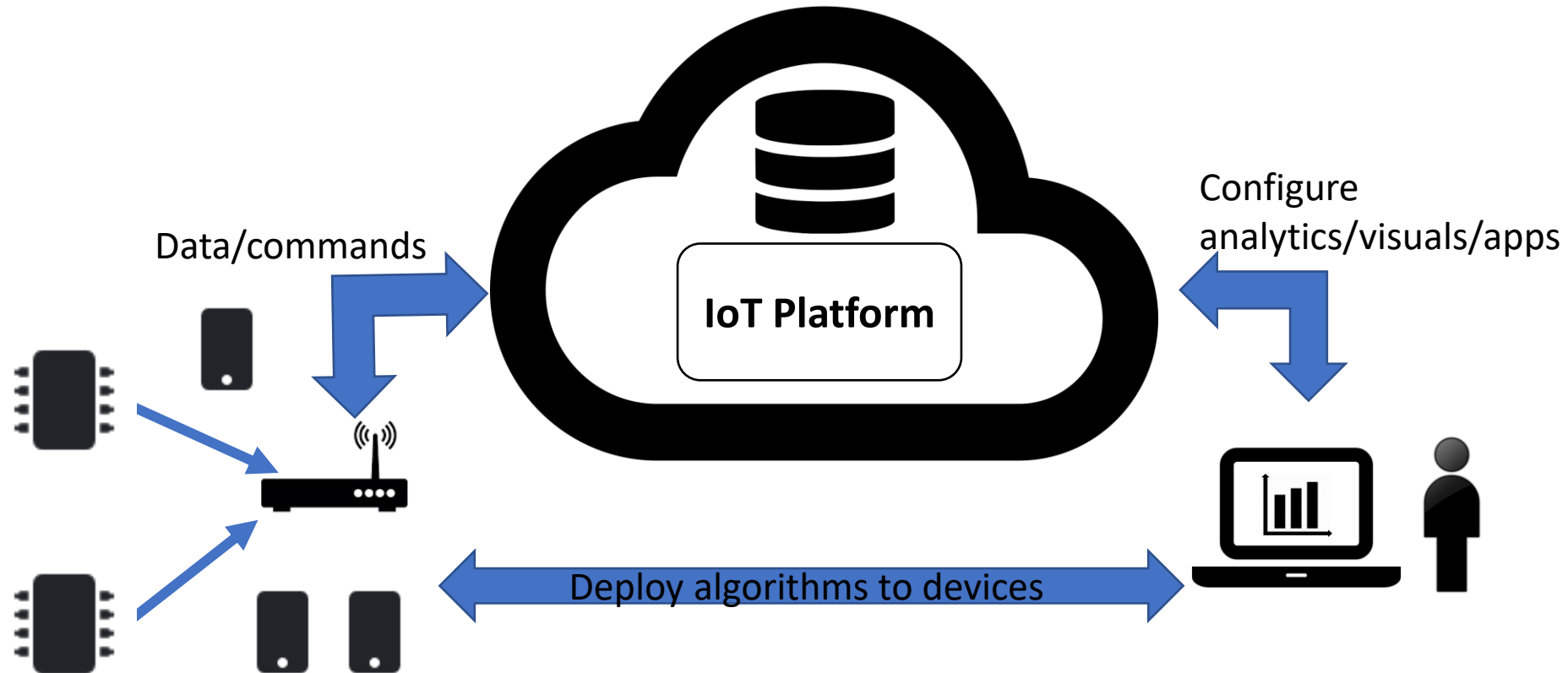
IoT Platform Characteristics

- Manage many concurrent device connections
- Connectivity across several connection types
- "Off-the-peg" IoT protocol stack
- Manage/analyse/visualise data
- Integrations to other services/apps
- App Development



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IoT Platform – general architecture



IoT Platform Advantages

- Use software component that has been pre-built and pre-tested. This increases the reliability of your application and reduces development effort.
- IoT frameworks constantly evolve, providing new features, integrations etc.
- Encourages better "design pattern" for your IoT app.
- Predefined APIs and docs
 - Great for collaboration
- "Baked-in" standards and features:
 - Security, authentication, scalability...

Which one?

- Connectivity
 - Does the platform provide suitable capability and integrations (WiFi/Cellular/LPWan-Sigfox)
- Maturity
 - In business for long? Critical mass in developer community?
- Free
 - Is there a free tier (handy for evaluation)?
- Service type
 - Platforms try to distinguish themselves – what specialisms/USP does it have?
- Security
 - What security model do they use? Is there security issues reported in past?
- Geographic area
 - Does it operate well at your location (can you select edges/data centres)





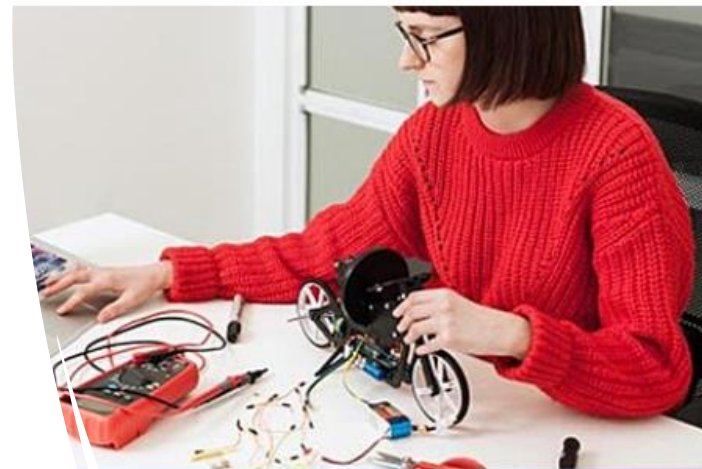
Examples:



Thingspeak

Thingspeak

- **ThingSpeak** is a cloud-based IoT platform to store and retrieve data from devices.
 - Uses HTTP protocol/Restful APIs
 - MQTT
- "Collect and analyse data quickly and easily"



Students and Educators

Projects quickly with built-in MATLAB data analysis tools and real-

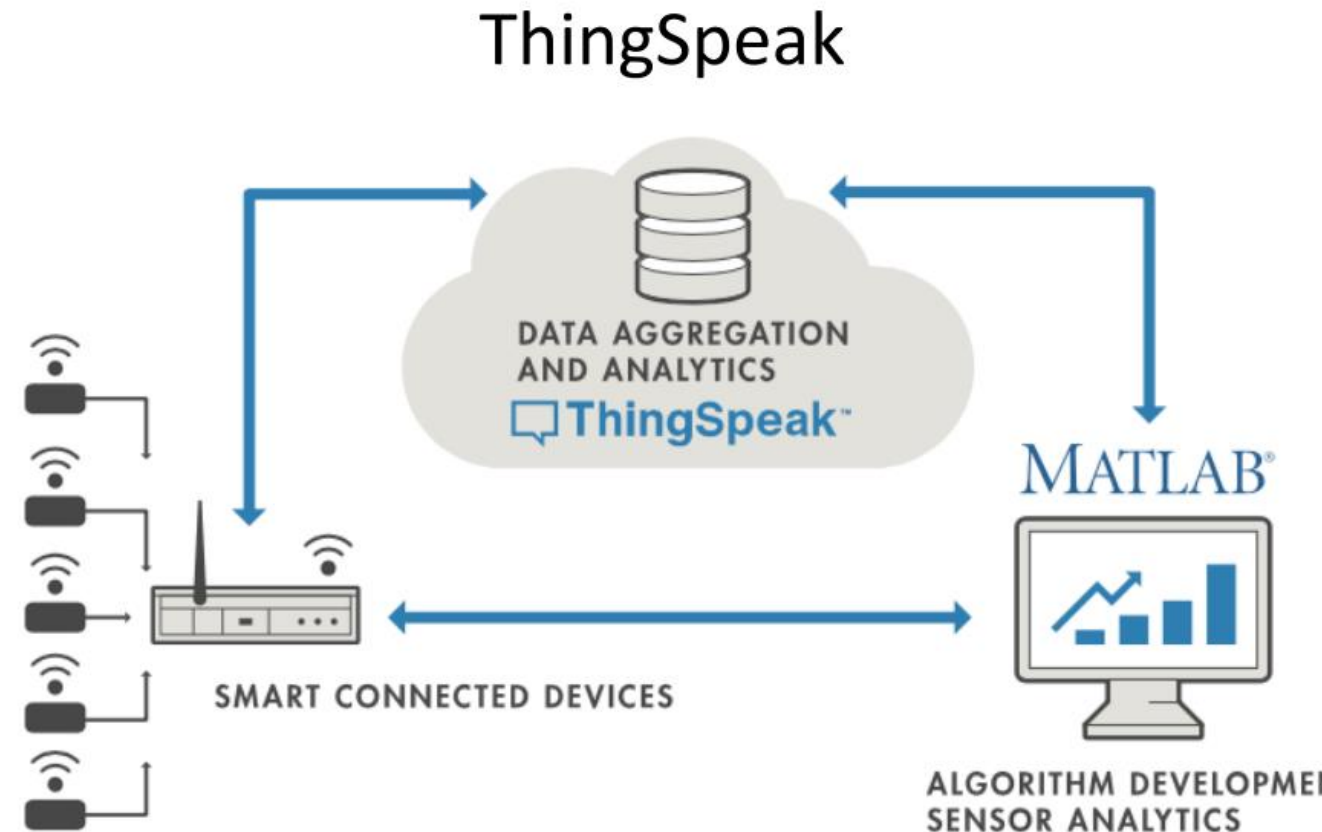


ThingSpeak for Environmental Monitoring

Build IoT services for remote monitoring of air quality sensors, and create MA to predict pollution levels

Thingspeak Overview

- Account-based
 - Can create free account online
- Brought to you by the people who made Matlab/Simulink
 - Uses Matlab features/toolboxes
- SDKs/librariys for popular languages/devices
- Restful API means should work with any device



Thingspeak – basic use

Create a new channel

- Channels collect data

Collect data in the channel

- Devices write data to channels

Analyse the data

- Run analytical algorithms/visualise your data

Act on the data

- Test for certain conditions and perform actions

ThingSpeak – Create new channel



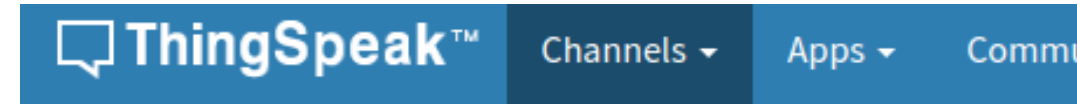
CREATE A NEW
CHANNEL TO COLLECT
DATA FROM DEVICES



DEFINE DATA FIELDS
FOR THE
CHANNEL(MAX 8)



CAN ALSO INPUT
LOCATION(LAT/LONG)
OF CHANNEL SOURCE



New Channel

Name	SensePi	
Description	Environment data from <u>senspi</u>	
Field 1	temperature	☒
Field 2	pressure	☒
Field 3	humidity	☒

Thingspeak - New channel

- Once saved you can access channel page:

SensePi

Channel ID: **625505**

Author: [fxwalsh](#)

Access: Private

Environment data from senspi

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

[+ Add Visualizations](#)

[+ Add Widgets](#)

[Export recent data](#)

[MATLAB Analysis](#)

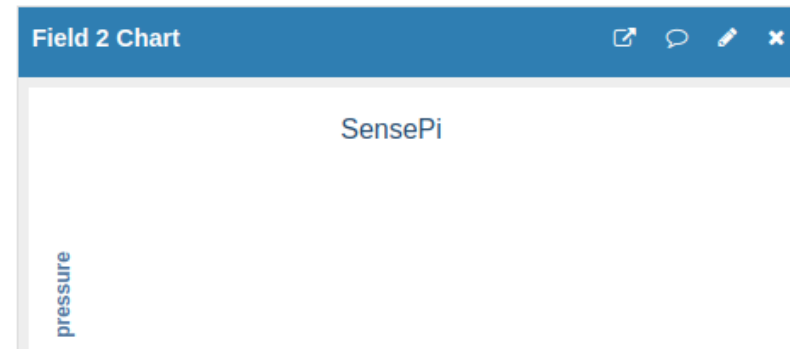
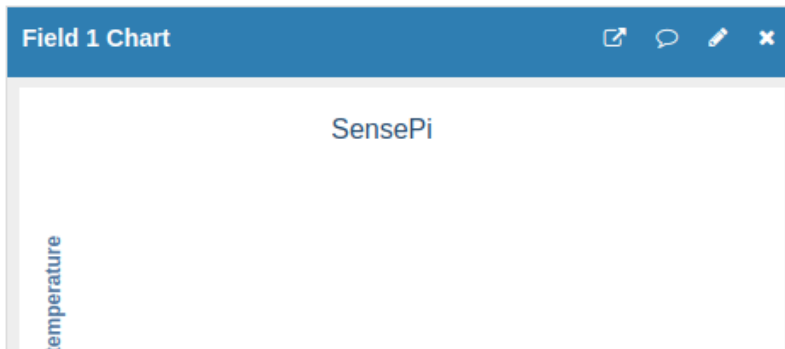
[MATLAB Visualization](#)

Channel Stats

Created: 4 minutes ago

Updated: 4 minutes ago

Entries: 0



Thingspeak - Add data to channel

- Programmatically, many ways!
 - Construct HTTP GET request and include field
 - MQTT (see in Lab)

API Reference

Use the REST and MQTT APIs to update ThingSpeak™ channels and to chart numeric data stc

ThingSpeak is an IoT platform that uses channels to store data sent from apps or devices. Wi
Using the REST API calls such as GET, POST, PUT, and DELETE, you can create a channel and
receive messages when a channel updates. Learn more about when to [Choose Between RES](#)

MATLAB® analysis and visualization apps enable you to explore and view your channel data.

REST API

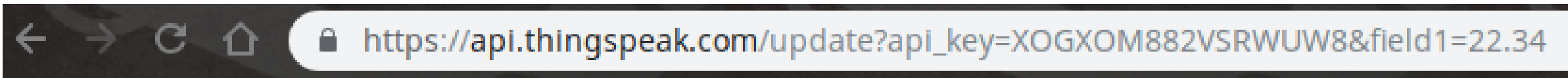
Use REST API calls to create and update ThingSpeak channels and charts

MQTT API

Use the MQTT API to update ThingSpeak channels

GET https://api.thingspeak.com/update?api_key=<WRITE-KEY>&field1=12

- Using HTTP GET request, can test from a browser:



Think Speak – Analyse data

- Thingspeak will visualise each field by default in channel view

SensePi

Channel ID: 625505

Environment data from senspi

Author: fxwalsh

Access: Private

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

+ Add Visualizations

+ Add Widgets

Export recent data

MATLAB Analysis

MATLAB Visualiz

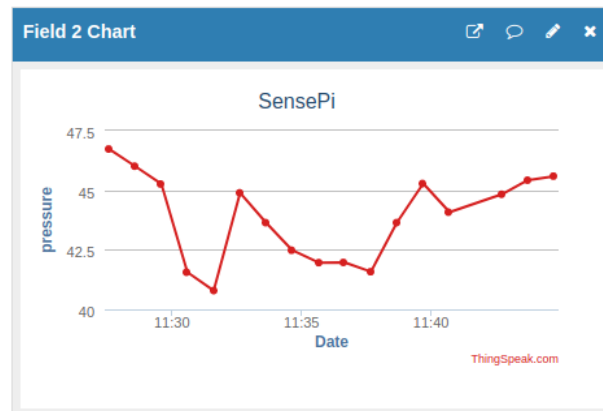
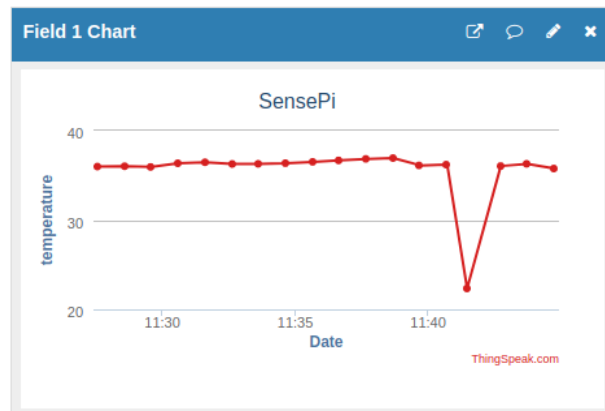
Channel Stats

Created: [about an hour ago](#)

Updated: [about an hour ago](#)

Last entry: [about a minute ago](#)

Entries: 17



Thingspeak - Apps

- The Apps tab provides various mechanism to transform, analyse, visualise and act on data.

Analytics



MATLAB Analysis

Explore and transform data.



MATLAB Visualizations

Visualize data in MATLAB plots.



Plugins

Display data in gauges, charts, or custom plugins.

Actions



ThingTweet

Connect a device to Twitter® and send alerts.



TimeControl

Automatically perform actions at predetermined times with ThingSpeak apps.



React

React when channel data meets certain conditions.

ThingSpeak Example: Analysis

- Can write Matlab Code to analyse and transform data
- Possible uses:
 - Clean data (remove outliers)
 - Statistical analysis
 - Transformations
 - Data Fusion
- Generally write results to second channel for further analysis/visualisation.

Thinkspeak: Convert Celcius to Fahrenheit

Convert temperature units

MATLAB Code

```
1
2 readChannelID = 12397;
3 % Temperature Field ID
4 temperatureFieldID = 4;
5
6 readAPIKey = '';
7
8 tempC = thingSpeakRead(readChannelID,'Fields',temperatureFieldID,'ReadKey',readAPIKey);
9
10 % Convert to Fahrenheit
11 tempF = tempC*1.8+32;
12 display(tempC,'Temperature in Fahrenheit');
13
14 % Replace the [] with channel ID to write data to:
15 writeChannelID = 1234;
16 % Enter the Write API Key between the '' below:
17 writeAPIKey = 'abcd';
18
19 thingSpeakWrite(writeChannelID,[tempF,tempC],'Writekey',writeAPIKey);
```

Save and Run

Save

Other Platforms

- Ubidots
- Amazon Web Services
- Microsoft Azure
- Evothings

Thingspeak – Add data with HTTP

- Make HTTP request from Python:

```
import requests

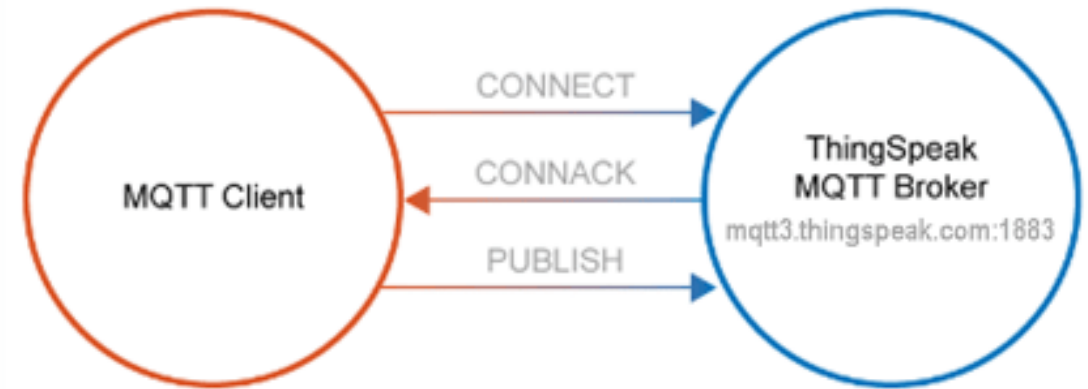
def writeData(value1, value2):
    api_key = "YOUR_WRITE_API_KEY"
    url = f"https://api.thingspeak.com/update?api_key={api_key}&field1={value1}&field2={value2}"

    response = requests.get(url)

    if response.status_code == 200 and response.text != '0':
        print("Data written successfully")
    else:
        print(f"Failed to write data. Response: {response.text}")
```

Thingspeak – Add data with MQTT

```
def publish_data_to_thingspeak(field1_value, field2_value):  
    # Thingspeak MQTT configuration  
    channel_id = "YOUR_CHANNEL_ID"  
    api_key = "YOUR_WRITE_API_KEY" # Your MQTT API Key  
    mqtt_host = "mqtt3.thingspeak.com"  
    topic = f"channels/{channel_id}/publish/{api_key}"  
  
    # MQTT client setup  
    client = mqtt.Client()  
    client.connect(mqtt_host, 1883, 60)  
  
    # Payload format: "field1=value1&field2=value2"  
    payload = f"field1={field1_value}&field2={field2_value}"  
  
    # Publish data  
    client.publish(topic, payload)  
    print(f"Published: {payload} to topic: {topic}")  
  
    # Allow time for message delivery  
    time.sleep(1)  
  
    # Disconnect from MQTT broker  
    client.disconnect()
```



1. Publish to a channel feed

channels/<channelID>/publish.

2. Publish to a channel field

channels/<channelID>/publish/fields/field<fieldnumber>

Which to use: HTTP or MQTT API

- HTTP (REST) API
 - You need to retrieve any historical data, such as data within a defined time range.
 - You want to Create Channel, Read Settings, Clear Channel, Delete Channel, or Create Chart.
 - You need a response!
- MQTT
 - Your device is power-constrained, and you want lower battery consumption to send data to ThingSpeak. Also, an MQTT PUBLISH operation is typically faster in this scenario.
 - Your device connectivity is intermittent, and you have limited bandwidth usage.
 - You want immediate updates of data posted to a channel.
 - You want messages pushed to you instead of needing to poll the server for new messages.